

Detecting Stability Near the Unit-Root Boundary

Research Question

How reliably can we detect changes in the stability of a time series system, and how does statistical inference behave under different conditions?

Focus: AR(1) processes close to the unit-root boundary

The AR(1) Model

$$X[t] = \text{beta}X[t-1] + \text{epsilon}[t]$$

Beta controls persistence: how much of the previous value can influences the current one.

Stable/ stationary: $|\text{beta}| < 1$

Unit root : Beta =1

Set up with ADF test

Fixed Hypothesis:

Null H_0 : $\beta = 1$ versus Alternative H_1 : $\beta < 1$

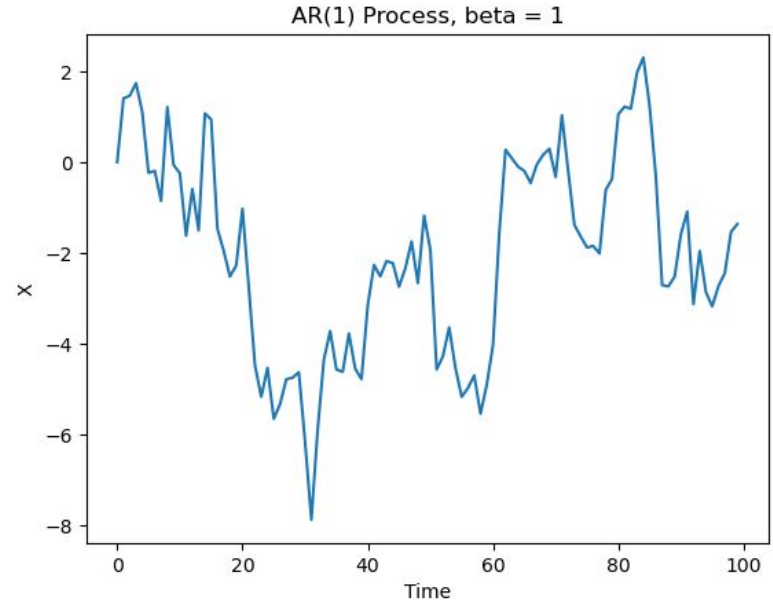
ADF test and the decision rule $p < 0.05 \Rightarrow \text{Reject } H_0$

First Experiment: Type 1 error

Fixed $\beta = 1 \Rightarrow H_0$ actually true

Ran ADF test repeatedly and got **$\alpha = 0.059$**

```
Number of rejections: 59  
Type I error rate: 0.059
```



Main Experiment: Power

Changed beta: 0.8, 0.9, 0.95, 0.99

With $T = 50, 100, 500, 1000$

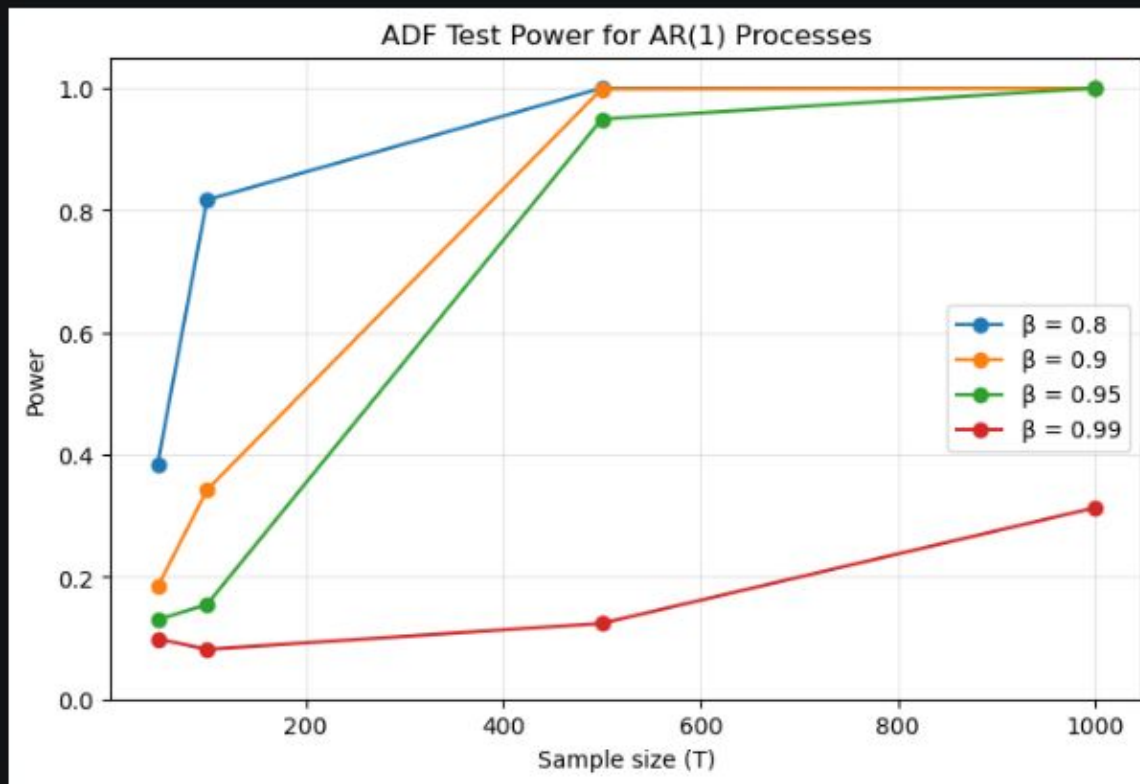
With 10,000 simulations for each condition

```
Beta: 0.8 T: 50 Power: 0.3845
Beta: 0.8 T: 100 Power: 0.817
Beta: 0.8 T: 500 Power: 1.0
Beta: 0.8 T: 1000 Power: 1.0
Beta: 0.9 T: 50 Power: 0.1834
Beta: 0.9 T: 100 Power: 0.3423
Beta: 0.9 T: 500 Power: 0.9986
Beta: 0.9 T: 1000 Power: 1.0
Beta: 0.95 T: 50 Power: 0.1298
Beta: 0.95 T: 100 Power: 0.1547
Beta: 0.95 T: 500 Power: 0.9493
Beta: 0.95 T: 1000 Power: 1.0
Beta: 0.99 T: 50 Power: 0.0984
Beta: 0.99 T: 100 Power: 0.0813
Beta: 0.99 T: 500 Power: 0.1238
Beta: 0.99 T: 1000 Power: 0.3132
```

[115]

✓ 0.1s

...



Beta = 0.99 Exp.

T = 50,100,150,200,300,400,500,750,1000

N_simulation = 5000

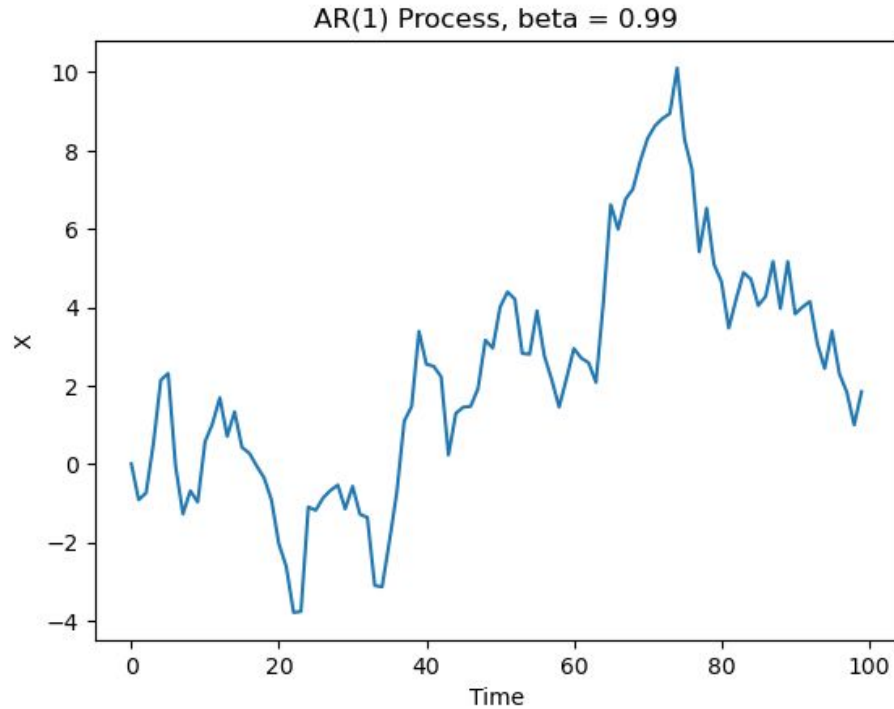
```
Beta: 0.99 T: 50 Power: 0.0988
Beta: 0.99 T: 100 Power: 0.0816
Beta: 0.99 T: 150 Power: 0.085
Beta: 0.99 T: 200 Power: 0.089
Beta: 0.99 T: 300 Power: 0.0888
Beta: 0.99 T: 400 Power: 0.097
Beta: 0.99 T: 500 Power: 0.1306
Beta: 0.99 T: 750 Power: 0.2022
Beta: 0.99 T: 1000 Power: 0.3144
```

RESULT

Conceptually we did expect: Power increases as T increases

But close to $\beta = 1 \Rightarrow$ at 0.99 it gets hard to detect even with bigger sample sizes.

The interesting result for me was that a process can be technically stable, $\beta = 0.99 < 1$, but still be extremely difficult for the ADF test to distinguish from a unit-root process.



Answer

Simulation shows that detecting stability becomes increasingly difficult as the AR(1) coefficient approaches the unit-root boundary $\beta=1$.

The ADF test performed reasonably well under the null, with a Type I error of about 5.9%, close to the nominal 5% level.

For stable processes, statistical power generally increased with sample size. However, when β was very close to 1, particularly $\beta=0.99$, the test had very low power even with relatively large samples. For example, at $T=500$, power was about 95% for $\beta=0.95$ but only 12% for $\beta=0.99$. At $T=1000$, power for $\beta=0.99$ was only about 31%.

Cont:

Thus, a system can be technically stable ($\beta < 1$) but extremely difficult to distinguish statistically from an unstable/unit-root system when it lies very close to the boundary. This demonstrates that the reliability of statistical inference depends strongly on both the distance from the stability boundary and the amount of available data

[Link to the code](#)

Other plots

When $\text{Beta} = 0.9$, the process tends to get pulled back towards 0.

